

Final Root-Zone Model vs Naive Baseline

48h unified model, no RH input - walk-forward test evaluation

Generated from scripts/exports CSVs

pH final model

MAE 0.330

R2 0.932

+62.4% MAE reduction

EC final model

MAE 0.127

R2 0.982

+35.3% MAE reduction

Detailed comparison on the same test rows

Naive baseline = carry forward the previous observed pH/EC value.

Target	Test SD	Final model	Naive baseline	Improvement
pH	1.605 observed	MAE 0.330 RMSE 0.414 (25.8% of SD) R2 0.932	MAE 0.877 RMSE 1.122 (69.9% of SD) R2 0.502	MAE -62.4% RMSE -63.1% N=57
EC	1.277 observed	MAE 0.127 RMSE 0.171 (13.4% of SD) R2 0.982	MAE 0.196 RMSE 0.531 (41.6% of SD) R2 0.824	MAE -35.3% RMSE -67.8% N=57

Scale check: RMSE relative to observed test-set variability

Lower is better. Bars show RMSE as a percentage of the measured SD.

